🎬 Movie Success Prediction and Sentiment Analysis Report

Abstract

In today's digital world, movies generate a large amount of data through ratings, reviews, and audience feedback. This project aims to predict whether a movie will be successful or not using machine learning techniques. Along with this, sentiment analysis is performed to understand audience opinions using movie descriptions. The TMDB 5000 Movies dataset is used for analysis. By applying Logistic Regression and VADER sentiment analysis, the project provides useful insights into how different factors influence a movie's success. This project highlights the importance of data science in real-world applications like the film industry.

Introduction

The success of a movie depends on many factors such as budget, popularity, ratings, and audience response. Traditionally, predicting movie success was based on experience and assumptions. However, with the growth of data science, we can now analyze large datasets and make predictions more accurately.

This project focuses on two main aspects:

1. Predicting movie success using machine learning

2. Understanding audience sentiment using text analysis

By combining these two approaches, we can better understand what makes a movie successful and how audience opinions impact performance.

Tools Used

- **Python** – For coding and analysis
- **Pandas & NumPy** – For data handling
- **Matplotlib & Seaborn** – For visualization
- **Scikit-learn** – For machine learning model
- **NLTK (VADER)** – For sentiment analysis
- **Google Colab** – For development
- **Dataset**: TMDB 5000 Movies Dataset

Steps Involved in Building the Project

1. Data Collection

The TMDB dataset containing movie details like budget, revenue, ratings, and overview was imported into the google colab.

2. Data Preprocessing

- Removed missing values

- Selected important columns

- Created a new column called success (1 = hit, 0 = flop) based on revenue and budget

3. Feature Engineering

- Extracted genres from JSON format

- Selected key features like popularity, ratings, and votes

4. Exploratory Data Analysis (EDA)

- Visualized data using graphs

- Observed relationships between budget, revenue, and popularity

- Identified trends in successful movies

5. Sentiment Analysis

- Used VADER to analyze movie overviews

- Converted text into sentiment scores

- Classified sentiment as Positive, Negative, or Neutral

6. Model Building

- Used Logistic Regression algorithm

- Trained the model using selected features

- Split data into training and testing sets

7. Model Evaluation

- Evaluated model performance using accuracy

- Achieved approximately 70–85% accuracy

8. Visualization

- Created graphs for sentiment distribution

- Compared success rates across genres

Conclusion

This project shows how data science can be used to predict movie success and understand audience opinions. The results indicate that factors like popularity, ratings, and sentiment play a major role in determining a movie's performance. Sentiment analysis helped in understanding how the tone of a movie description affects its success. Overall, this project demonstrates the practical use of machine learning and natural language processing in the entertainment industry. It also provides valuable insights that can help producers and analysts make better decisions.